## TOPIC 1 Linear Equations and Inequalities

Name _____ Date _____

**1.** What value of $p$ makes the equation $7 - 3(p + 2) = 10 - (2p + 5)$ true?

**F.** −4

**G.** 4

**H.** −6

**J.** No solution

**2.** Solve the equation shown.
$0.5(4x - 6) = -1.5x + 4$

**3.** Does the equation shown below have one solution, no solution, or an infinite number of solutions? Explain your reasoning.
$\frac{1}{2}(8x + 6) = 2(2x + 1) + 2$

**4.** A triangle has a height of $(x + 5)$ inches and a base of 6 inches. A rectangle has a length of 4 inches and a width of $(x - 1)$ inches. The area of the triangle equals the area of the rectangle. What is the value of $x$?